

RESEARCH INTERESTS

Reinforcement Learning, Multi-armed Bandits, Uncertainty Quantification

EDUCATION

Indiana University <i>Ph.D. in Computer Science</i>	Bloomington, IN <i>Dec 2027</i>
Indiana University <i>M.S. in Data Science</i>	Bloomington, IN <i>May 2021</i>
West Bengal University of Technology <i>B.Tech. in Computer Science and Engineering</i>	Kolkata, India <i>June 2016</i>

PUBLICATIONS

- (C1) **Palash Chatterjee**, Roni Khardon. Improving planning and MBRL with temporally-extended actions. *Conference on Neural Information Processing Systems (NeurIPS) 2025*.
- (C2) **Palash Chatterjee**, Ashutosh Chapagain, Weizhe Chen, Roni Khardon. DiSProD: Differentiable Symbolic Propagation of Distributions for Planning. *International Joint Conference on Artificial Intelligence (IJCAI) 2023*.
- (W1) **Palash Chatterjee**, Roni Khardon. Planning with temporally-extended actions. *Workshop on Bridging the Gap Between AI Planning and Reinforcement Learning, AAAI Conference on Artificial Intelligence 2025*.

EXPERIENCE

Indiana University <i>Research Assistant</i>	Bloomington, IN <i>Aug 2020 - Present</i>
Currently working with Dr. Roni Khardon on improving planning and learning capabilities in model-based reinforcement learning.	
Indiana University <i>Data Analyst</i>	Bloomington, IN <i>Nov 2019 - Aug 2020</i>
Worked with Dr. Michele Roberts to analyze the relationship between K-12 education in Indiana and subsequent performance of the students in computer science courses.	
ThoughtWorks Technologies <i>Application Developer</i>	Gurgaon, India <i>July 2016 - July 2019</i>
Migrated existing MR pipelines and built custom Spark pipelines for data ingestion, cleanup and transformations to predict after-sales service.	
Built Jenkins pipelines to enable continuous integration and deployment of code in various environments.	
Led a team of 5 to develop a proof-of-concept for predicting equipment based on after-sales invoice with an accuracy of over 70% using decision trees.	

PROJECTS

SympyToTorch

Developed an utility to generate a PyTorch computation graph for a SymPy functions.

Using IMPALA as rollout policy for Monte Carlo Tree Search

Implemented single learner IMPALA architecture and used the same as rollout policy with Monte Carlo Tree Search.

Outlier detection using C2C-Siamese Networks

Detect outliers, by comparing the difference in feature representations of classes, using a Siamese Network, with an accuracy of 70% on MNIST.

SERVICE

PC Member

AAAI 2024, 2025, 2026
ICRA 2026

TEACHING EXPERIENCE

Indiana University B659: Topics in AI: Reinforcement Learning	Bloomington, IN <i>Spring 2022, 2024, 2025</i>
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HONORS AND AWARDS

Winner of International Planning Competition - Probabilistic Track

2023

International Conference on Automated Planning and Scheduling

Variant of DiSProD for planning in domains involving continuous and discrete variables, with dynamic selection of hyper-parameters.

Luddy Outstanding Research Award

2021

Luddy School of Informatics, Computing and Engineering, Indiana University

TECHNICAL SKILLS

Proficient in Python, PyTorch, JAX, Numpy, Java, SQL. Familiar with MapReduce, Spark, Hive, C++.